HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered **Java Spring Boot** REST APIs for inter-service communication, handling approximately **1,500 requests per second** during peak loads for enterprise backend systems.
- Developed and maintained complex data extraction and transformation logic using **Python**, enabling automated reporting that reduced manual effort by an estimated **20%** for operational teams.
- Managed production incident response for critical enterprise Java/.NET applications, resolving P1/P2 issues within defined SLAs and coordinating root cause analysis using **Splunk** and **Dynatrace**.
- Collaborated with cross-functional software teams to integrate drivers and peripheral software, contributing to the stability of core platform releases.
- Designed and implemented scalable build system software and tools to generate various image flavors for embedded systems, improving build efficiency.
- Performed image and system-level sanity testing to ensure software stability and feature completeness before release.
- Containerized microservices using **Docker** and deployed them to **AWS EC2** instances, managing configurations via **Git** and Jenkins pipelines.

SKILLS

Languages – C, C++, Python, Assembly, SQL, Java, Bash

Embedded Systems – RTOS, Device Drivers, Integration, Debugging, Zephyr RTOS

Tools & Platforms – JTAG, Git, Gerrit, GitHub, Perforce, SCons, CMake, Jenkins, AWS

Operating Systems – Linux, Real-Time Operating Systems (RTOS)

Methodologies – Agile, Scrum, Software Architecture, CI/CD

Cloud – AWS EC2, S3, Lambda, CloudWatch

PROJECTS

Embedded Linux System Integration Framework

- Developed a Python-based framework for automating the integration of device drivers and peripheral software onto embedded Linux platforms, streamlining the testing process.
- Implemented build system tools using **CMake** to manage dependencies and configurations for diverse hardware targets, reducing build times by **15%**.
- Integrated **JTAG** debugging capabilities into the framework, enabling engineers to perform low-level system debugging and root cause analysis more efficiently.

RTOS Kernel Module Development & Testing

- Engineered C/C++ kernel modules for a real-time operating system (RTOS) to manage critical hardware interfaces and optimize task scheduling for performance-sensitive applications.
- Designed and executed comprehensive unit and integration tests for RTOS components, achieving over **90% code coverage** and identifying race conditions early.
- Contributed to the development of a **Zephyr RTOS**-based application, focusing on power management features and ensuring adherence to strict real-time performance KPIs.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)